Leidenschaft
fürs Leben.

Guidelines for Academic Writing, Scientific Papers & Theses

SRH Hochschule Heidelberg



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1 Preface

Any sort of academic writing at SRH - within the Faculty of Business – be it a (Bachelor or Master) thesis, a research or internship report (and at times even a powerpoint presentation) – should adhere to internationally recognized standards. This document provides guidance on these standards, so that both students and supervisors have a common basis for the writing and evaluation of academic texts. As such, however, it is not neither an all-inclusive compendium covering each and every theoretically possible combination of (scientific) text, topic, form or occasion nor could it properly account for the different educational levels (ie Bachelor versus Master). Hence, next to this document, it is any author´s obligation and the responsibility of any SRH student to consult both additional external sources on "How to" as well as his/her supervisor, if in doubt.

Any deviation from these guidelines or (to be) known international standards in academic writing are to be avoided by all means as these may impact the evaluation result. Authors should simply keep in mind that the purpose of any such document is to demonstrate own proficiency and appropriate written academic communication. It shows the ability to create new knowledge by applying information retrieval skills, critical and analytical thinking, problem solving and persistence in continuous reading. Especially one´s thesis at the end of either Bachelor or Master education has to be understood as business card and entry into both job and career opportunities.

Nevertheless, especially for reports or presentations without scientific/academic claim, deviations might be well justified, sometimes even obvious or simply desirable. In any such cases, of course, other formats and standards are more appropriate and fine.

2 Structure of Scientific Papers

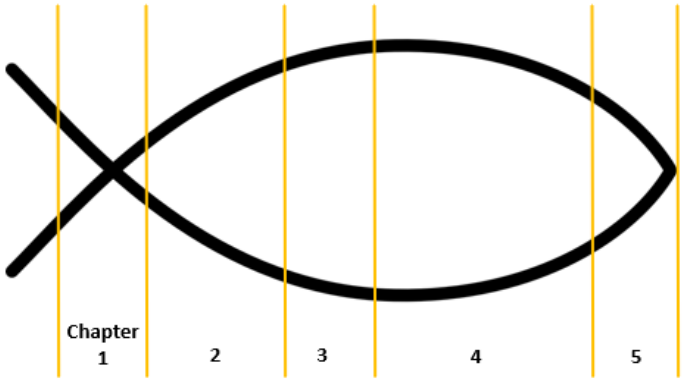
In general, the overall structure of the complete paper is as follows:

- Title page
- (Preface / Acknowledgement – if applicable/desired)
- Abstract
- Table of Contents
- (List of Abbreviations – if applicable)
- (List of Figures – if applicable)
- (List of Tables – if applicable)
- Main Body
- Reference List / Bibliography
- (Appendix – if applicable)
- (Declaration of Honor – if applicable/desired)

Focusing on the main body of any academic research paper, it is vital to understand and apply the following 5 chapter structure:

1) Introduction
2) Literature Review
3) Methodology
4) Analysis
5) Conclusion

The following diagram illustrates how much volume (eg by means of word-count and/or page numbers) each chapter should – roughly – contain:



The above represents an international, well-recognized standard. Deviations, although possible, are to be avoided and simply run the risk of causing misinterpretation. Rare exceptions to this structure might apply. Especially if no data is collected chapter 3 becomes obsolete and might be skipped. Nevertheless, please be aware that any such deviation has to be well justified and obvious to any third party. (The remark that a supervisor was fine with an alternative structure is as such no acceptable justification. Amongst others, because it would not be obvious to any third party.)

The following sub-sections detail the 5 chapters mentioned above. Please consult Chad Perry`s (1994) "A STRUCTURED APPROACH TO PRESENTING Phd THESES: NOTES FOR CANDIDATES AND THEIR SUPERVISORS" in order to complement and deepen the following introduction to the 5 chapter structure.

2.1 Introduction

In the first chapter of an academic paper, the author introduces the topic and problem to solve. An introduction usually comprises four parts. First, the problem to solve or question to answer is introduced. The relevance and importance of the problem should be motivated. Presenting the problem legitimizes the work, i.e. shows that there is a real need for a solution to a specific problem and a knowledge gap in research. Second, the objective and purpose of the paper should be stated. A research paper can have one or more objectives. The hierarchy between the different objectives should be clearly stated. In this part, it can also help to explain the target group for which the work is relevant and why. This is also the place to narrow down the topic in terms of limitations (ie what will not be answered). This part is followed by an explanation of the course of the investigation. The author explains how the papers objective is reached. The methodology as well as used data and origin of data can be briefly explained. Lastly, the structure of the paper should briefly be described.

In addition, chapter 1 shall cover – maybe as separate sub-chapters, depending on report/thesis content and overall length:

- Background to the Research and Research Justification
- Research Problem

2.2 Literature Review

The Literature Review is a thorough review of existing literature on the research topic. It is an account of what has been published on a topic by accredited(!) scholars and researchers – mainly in high-rated journals and partly in academic textbooks. It provides definitions for key terms and concepts. It summarizes (and on Master-level even more so) paraphrases, critically analyses and evaluates previous research findings available on a subject and presents them in an organized way. Past research findings should be discussed to highlight existing research gaps. Also, the main areas of agreement or controversy in the research field can be discussed. A good literature review demonstrates an in-depth understanding of the topic including key concepts, terminology, theories and definitions as well as corresponding authors. However, it is insufficient to merely string together what other researchers have found. A decently critical reflection and discussion of the body of knowledge is the ultimate goal of this chapter. Hence, it is especially important to base chapter 2 on a decent amount of A-rated journal articles.

2.3 Methodology

This is a crucial chapter in any thesis since the validity of the study will to a large extent depend on the care taken in the planning and implementation of the research. The research design should stand up to the test of 'replication' by other competent researchers. Any conclusions drawn depend largely on the validity and accuracy of the collected and analyzed research data, so great care should be taken both to describe the research design and to outline the approach to data analysis. Within this chapter, no attempt to offer any findings of the research should be made - outline the research design only.

This chapter describes how the research question will be answered. For descriptive papers it usually describes the theoretical templates used. For empirical papers it describes the research design, population, data used, sampling techniques, data analysis models and limitations of methodology. (On Master-level it might include a brief discussion on epistemology and ontology – then being part of the first sub-chapter, below.) The following subchapters give a brief overview about the relevant content to be covered per chapter.

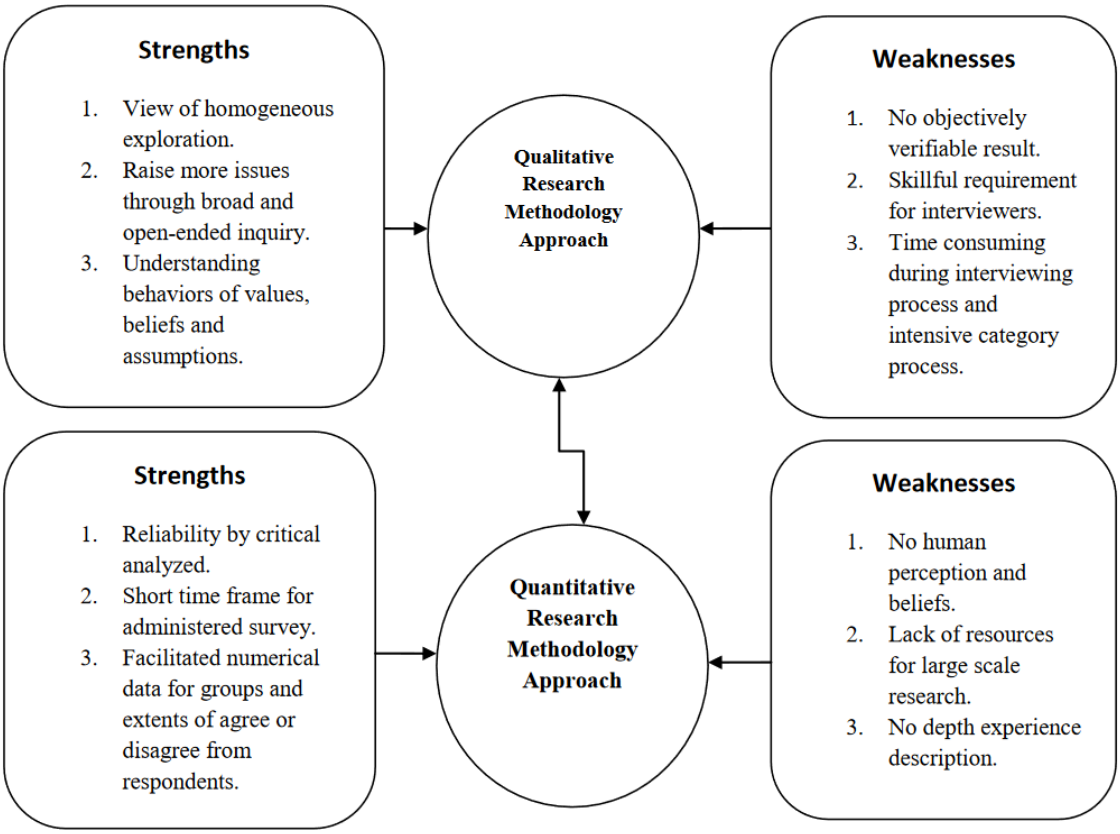
2.3.1 Research Paradigm

The research question can be subject to a qualitative or quantitative approach – or a mix of both. In addition, conclusion drawing can be based on inductive or deductive reasoning. Chapter 3 has to provide the reader with a clear yet reflective and critical, theory-based explanation of

- the method(s) used to collect data
- why it was collected this way (and not in any other way)
- drawbacks of the collect method(s) (and how these were taken into consideration)
- and, finally, how the collected data was analyzed

By extensively studying the literature it becomes evident which of the above is to be used.

Below a comparing overview of strength and weaknesses of each:



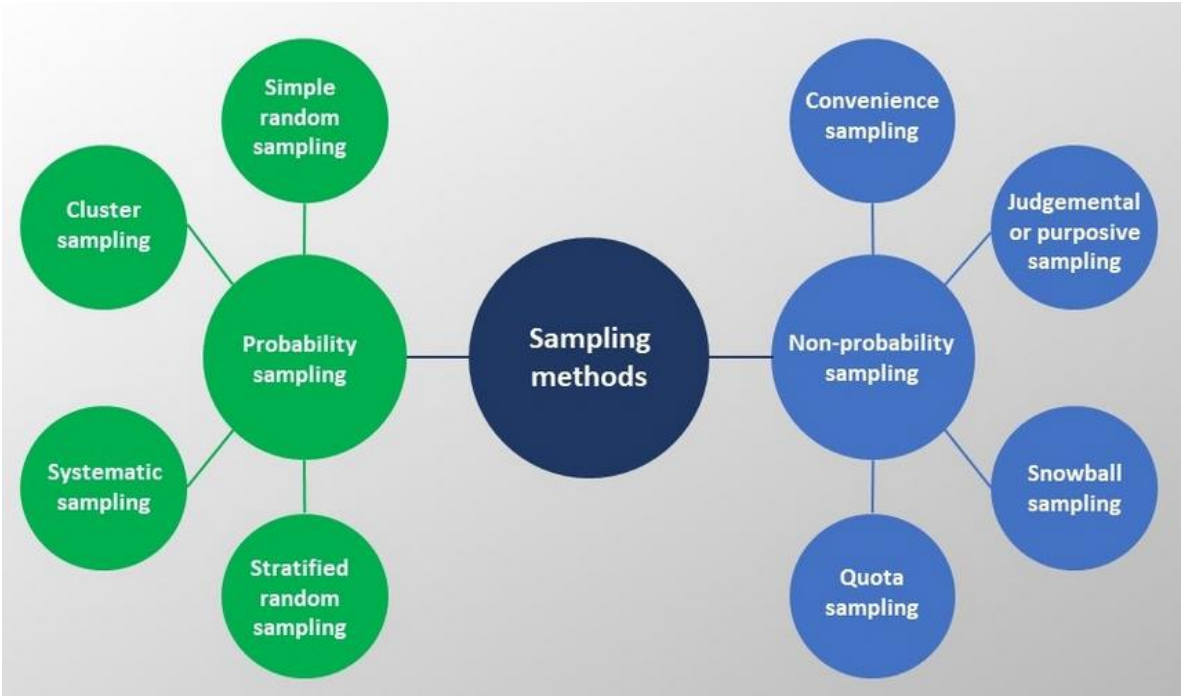
Source: "The Strengths and Weaknesses of Research Methodology: Comparison and Complimentary between Qualitative and Quantitative Approaches" (p. 101) by Looi Theam Choy / Journal Of Humanities And Social Science, Volume 19, Issue 4, Ver. III (Apr. 2014), PP 99-104

2.3.2 Research Design

The researcher must select a research design that is appropriate for the topic being investigated. There is an array of different designs from which to choose and each of these might need to be discussed. Once the most suitable design has been selected, much more reading of literature pertaining to the selected research design is required. This section is the most important one within chapter 3. It specifies the applied methods and methodology used to collect data. On top, it in-forms the reader about pros and cons of the chosen approach(es).

2.3.3 Population / Sampling

Since it is hardly possible to investigate a whole population, a sample has to suffice. If data is collected the sampling strategy has to be laid open and (critically) discussed. In essence, there exist probability and non-probability sampling strategies. Probability sampling provides every member of the population with the same chance of being selected. It is mainly used in quantitative research and produces results which are representative of the whole population. In a non-probability sample, individuals are selected based on non-random criteria. Hence, not every individual has a chance of being included. This type of sampling has a higher risk of sampling bias and any inferences about the population are weaker than with probability samples. Therefore, conclusions may be more limited. Non-probability sampling techniques are often used in exploratory and qualitative research. The following diagram illustrates the four most important ones within each category:



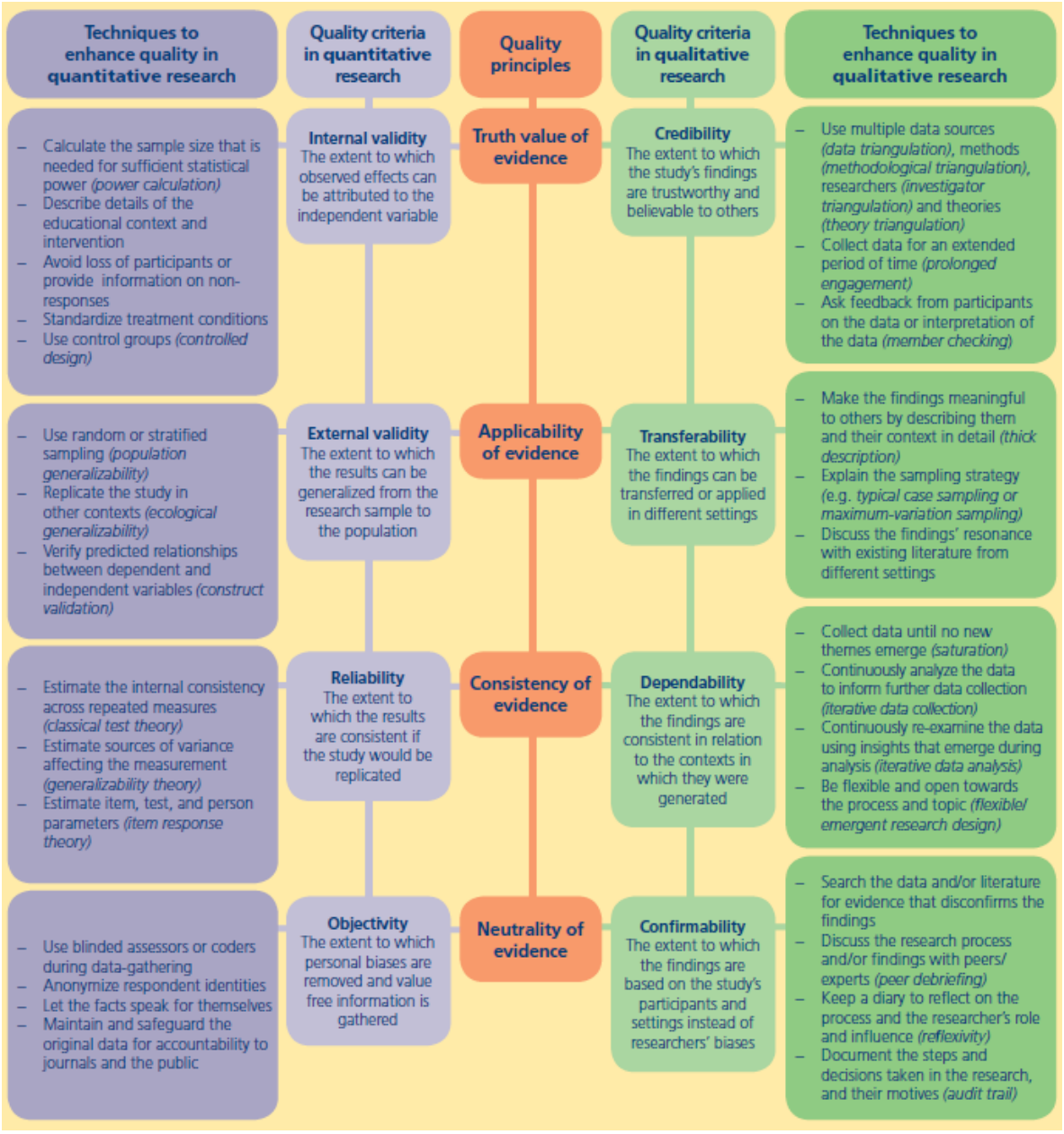
Please consider also: "The use of sampling methods in advertising research: a gap between theory and practice" by Sarstedta, Bengartc, Shaltonid & Lehmann / International Journal of Advertising, June 2017.
A primer on the pros and cons of these different sampling methods can be found within: "Pros and cons of different sampling techniques" by Gaganpreet Sharma / International Journal of Applied Research, 2017; 3(7): 749 - 752

In addition, the reader shall be informed about both the unit of analysis as well as the unit of observation. The former will be determined by the research question; the latter is determined largely by the method of data collection that used to answer that research question. The first step on how to analyze data is to define a unit of analysis, which is the "who" or the "what" that you are analyzing for your study, e.g., an individual student, a group, or even an entire organization. It is the entity the researcher is focusing on. A unit of observation is the item (or items) that you observe, measure, or collect while trying to learn something about your unit of analysis.

2.3.4 (Minimum) Scientific Quality Criteria

Good quality research provides robust evidence, has been conducted ethically, stands up to scrutiny and is informative towards the scientific community. Practically spoken, research should lead to some new idea and the credibility or the usefulness of this new idea shall be corroborated by theoretical or empirical arguments. This is – amongst other things – achieved by the demand for transparency and repeatability of any conducted research.
For a better understanding of what constitutes quality research, please be referred to "Evaluating research: A multidisciplinary approach to assessing research practice and quality" by Mårtenssona, Forsb, Wallinc, Zanderd & Nilssone, Research Policy, 45 (2016), 593–603.
There are minimum criteria to evaluate the quality of scientific research stemming from measurement theory. Any scientific work shall adhere to at least the principles presented in the following diagram. However, depending upon beforementioned difference of qualitative versus quantitative research it becomes obvious that what constitutes good quantitative research might well differ from good qualitative research. Hence, different criteria apply.

A very helpful overview is provided by Frambach, van der Vleuten, & Durning (2013):



Academic Medicine: April 2013 - Volume 88 - Issue 4 - p 552

When digging deeper, Kaya Yilmaz´ "Comparison of Quantitative and Qualitative Research Traditions: epistemological, theoretical, and methodological differences" (European Journal of Education, Vol. 48, No. 2, 2013) will provide you with important insights as well as with an excellent reference list for further reading.

2.4 Analysis

This chapter provides an in-depth analysis of the collected data. It is the most important chapter of any thesis/report. Results describe what you discovered / observed. Discussion explains the findings and relates them to previous research and the research question. Hence, this chapter answers the question "so what?" It contains a detailed description of the results obtained and their interpretation. Ideally, these results are reflected towards both empirical results from other researchers as well as the discussed literature in chapter 2. It is important to (tightly) link the results of chapter 4 with the content of chapter 2.

2.5 Conclusion

The final chapter summarizes the main findings and answers the central research question(s). In addition, the author should distance him/herself from his/her own work to show he/she can critically and realistically evaluate the research approach and findings. Furthermore, existing research gaps as well as an outlook on possible future developments complements the Conclusion by means of recommendations to guide future research.
A jigsaw puzzle analogy might be useful for understanding what chapter 5 is about: Research begins like a jumbled jigsaw puzzle about the research problem. Chapter 2´s literature review starts putting the pieces together to try to uncover a picture, but shows that some pieces are missing and so the complete picture cannot be known. Then chapter 3 and 4 describe the hunt for the missing pieces and the matching together of a few newly found pieces. Finally, chapter 5 returns to the puzzle, briefly summarizing what the picture looked like at the end of chapter 2 and then explaining ow the new and old pieces fit to make the whole picture clear.

Reminder: Please consult Chad Perry`s (1994) "A STRUCTURED APPROACH TO PRESENTING Phd THESES: NOTES FOR CANDIDATES AND THEIR SUPERVISORS" in order to complement and deepen the 5 chapter structure.

3 Language Rules for Scientific Research Papers

Apart from structure and content, language plays an important role in defining the quality of an academic paper. To ensure that the linguistic level is adequate there are certain rules to follow, which are listed below.

3.1 Central Linguistic Rules

- Targeted use of both punctuation and paragraphs can strengthen the desired message for the reader. Paragraphs consisting of only one sentence are inappropriate.
- Long, elaborate sentences should be avoided. A high language level should not obstruct meaning. People who can take complicated ideas and write them simply show that they fully comprehend the meaning of what is being written about.
- Using professional terminology used in the research field raises the quality of a paper. Professional terminology should be used to precise not complicate the text.
- When using professional terminology specific to the subject discussed in an academic paper, be cautious in regard to interdisciplinary differences in meaning. Consider context: does one speak in a sports club about members, customers, consumers, etc.? If a term leaves room for interpretation goes, it requires a precise explanation.
- Superfluous vocabulary, or "fluff", should be avoided. Sentences should be precise and focused statements.
- Slang and platitudes need to be avoided.
- Arguments and claims should be supported by citing (empirical) studies, books, etc. The author's own opinions should be used sparingly and only if the argumentation is made plausible. Personal opinions should not be expressed in first-term, using the "I"-form. Ideally, an author or editor should not be spoken of ("In the author's opinion…."). Instead, the use of neutral and objective statements is preferred ("It is recommended…" or "One should…."). This is not intended to be contradictory to a critical discussion of a clearly identifiable author-ego, theoretical models, third party statements, empirical realities, etc.

3.2 Additional Language Rules

- The first mentioning of an organization can be accompanied by a short description either in the text or in a footnote (eg, "the international market research institute, Repucom…").
- Quotation marks cannot only be used for citations or for definitions, but can also be used to place distance from specific text/words.
- Sometimes it is wise to provide qualifications or background of an author when first mentioned in a text. It contributes to the clarity of the text, grants authority of the person being referenced, and classifies their statements in context of the paper ("Luhmann is considered one of the fathers of systems theory…").

4 Formal Rules for Scientific Research Papers

The guidelines listed below are based on the American Psychological Association (APA) standards. "APA style" receives continuous updates. Hence, it is mandatory to consult publicly available sources to understand and apply latest changes within any thesis. However, a certain degree of freedom is acknowledged in case of sticking to any older version.

4.1 Layout and Design

4.1.1 Overall Structure of an Academic Research Paper

The formal layout of a research paper is divided into separate parts:

- Title Page (centered on the page)
- Confidentiality Clause (if included, this should be clearly marked on the cover page)
- Abstract
- (Potential) Foreword
- Indexes (table of contents, list of abbreviations, list of figures, list of tables)
- Main text body
- Bibliography
- Potential appendices
- Declaration of honor
- Enclosures (can include things like a USB-stick in an attached pocket)

The points above do not correspond to the chapters of the research paper. Any forewords, title pages, and directories are included without numbering in the table of contents, but they do require page numbers using Roman numerals. The first chapter begins on arabic page one. The appendices and bibliography are also included in the table of contents without numbering and Roman page numbers. The Declaration of honor should not be included in the table of contents and put at the very end of the paper.

4.1.2 Title Page

The cover page must include: next to the desired degree and name of the institution (university and department) the school's logo on the upper third of the page; title of the work at the lower end of the top third of the page; name of the supervisor, author's contact information, student matriculation number, and the date of submission on the bottom half of the page.

4.1.3 Table of Contents

The table of contents shows the complete structure of the research paper (including page numbers) by use of a decimal classification system. It shows the structure and content of the work. When a research paper is evaluated, the table of contents plays a critical role in determining the quality of the work. It is often the case that a good outline indicates a good qualitative treatment of the topic being discussed. The most important aspects when creating a table of contents are listed below:

- Main headings and subheadings are formatted clearly
- Maximum of 3 to 4 levels of subheadings
- At least two subheadings per main heading
- Similar weight and detail of each subheading
- Specific, concise and content-neutral headings, no formulation of "action titles", no questions as headings
- No unknown formulas or symbols
- Simple identification of page numbers (no indication of page spans, e.g. 1-8); aligned on the right side

Example of a single chapter:

1 INTRODUCTION 1

1.1 Introduction 1

1.2 Background to the Research & Research Justification 5

1.3 Research Problem 6

1.4 Outline of this Thesis 7

1.5 Theoretical Framework & Limitations 10

1.5.1 Costs 11

1.5.2 Motivational Issues 11

1.5.3 Cultural Issues 13

1.6 Conclusion 14

4.1.4 Page Formatting

For the appearance of the research paper, use the following guidelines:

- **Page Format:** DIN A4, single-sided
- **Text Format:** size 12 point, justified with hyphenation
- **Font:** Arial
- **Spacing:** 1.5 lines (not applicable to bibliography which requires single-spacing)
- **Pagination:** No pagination for the title page. Roman numerals from the confidentiality clause up to the main text body. Starting with the first page of the first chapter, use Arabic numerals either above or below in the right corner. After the main text body, starting with the bibliography, use roman numerals again.
- **Highlighting:** use Italics (but sparingly)
- **Margins:** right, left, top and bottom: 2.5 cm

4.1.5 Headings

Different level headings require different formatting:

- Top Heading: size 14 point, bold
- Second Heading: size 12 point, bold
- Third Heading: 12 point, normal
- Fourth Heading: 12 point, Italic

Please note that after the last number of a chapter number there is no decimal/period at the end. Use fourth-level headings sparingly. All headers have to be numbered. Each header-level has to appear at least twice within the same sub-structure. The following example shall provide clarity on this:

1	1
1.1	1.1
1.2	2.
2	3
correct	incorrect

4.1.6 Footnotes

Footnotes should be numbered consecutively within the text immediately following a word in size 10-point superscript font. The text of the footnote should also be size 10-point, single-spacing at the bottom of the same page where the footnote appears. If a footnote is for a single word, it should be immediately after that word. If the footnote refers to the whole sentence, place the footnote at after the period.

4.1.7 Cross References

When cross-referencing within a paper, be sure to use a chapter and/or section (like an Appendix) instead of just a page number.

4.2 Citations and References

4.2.1 General Rules for Citations

"Mine or yours?" This question is highly relevant for scientific work. If ideas or statements from others are in any way implied directly or indirectly in a text, they must be cited properly. If not, this results in a "deadly sin" against recognition of research in scholarly work. In certain cases of missing or inadequate citation, legal prosecution – due to plagiarism - may follow. But even in minor cases, inappropriately acknowledging the source will result in lower grades.

Correct citation is essential in academic work and not only proves the origin of a statement or idea but also documents that the writer is familiar with the literature. References within a paper are usually made either in-text or as a footnote, indicating clearly how to find the complete source documentation in the bibliography.

There are two possible methods for citations: the Harvard method (of which APA-style is a widely used variation), where the citation is made in the text in parentheses immediately following the quote or paraphrased text after the period. The alternative method is using footnotes. The citation will then be indicated in-text with a superscript, with the footnote at the end of the page.

Although you enjoy the freedom of choice – either Harvard, APA or footnotes: It is of utmost importance to follow your preferred way of referencing consistently throughout the paper. In addition, please consult the official referencing style guidelines in order to include the very latest

versions applicable to your thesis/report. In either case, it vital to include page-numbers when referencing!

4.2.2 Indirect Citation

When dealing with indirect quotations, it is important to establish a clear relationship between the quoted text and source, so the reader doesn't have to laboriously differentiate between one's own opinion and another's. So, if only one part of a sentence, or the whole sentence, or a paragraph refers to a source then the citation should be after the respective section and not just anywhere in the text. There are three ways to cite paraphrased or summarized information in a thesis:

- The source is cited at the end of the paraphrased section. Example: Compared with the meaning from the literal translation, this quote can instead be used as the best explanation of ambiguities (Kramer, 1995, p. 146).
- The author is mentioned in the text and the year and page are included in brackets immediately after. Example: Even Kramer (1995, p. 146) believes that…
- The author(s) and the year are part of the text. Example: As early as 1995, Kramer (p. 146) points out in his article that…

4.2.3 Direct Citation

Direct quotes present another author's original statements. Direct quotes shall support or illustrate the author's argumentation, they are not intended to replace it. They are either integrated into the text of a research paper or set apart as a block. A direct quote is framed with double quotation marks (when the quote contains another direct quote, use single quotation marks). Be sure to cite the source of the quote at the end of the sentence/section either directly or within a footnote. Direct citation within the text immediately follows the quotation and includes the author's last name, year of publication, and page number(s) with commas in between (all in brackets): "(Last Name, Year, Page Number)".

> Example: "With so much 'scientific' work, however, suspicions are raised so high that dealing with literary documents is dubiously termed 'impression management' nowadays" (Nitsch et al., 1994, p. 65).

Direct quotes of 30 words or more must be visibly indented as a block text (see example below). The quote goes down to a new line and is set 1 cm in from both left and right sides; also, there are no quotation marks around the quote, unless it includes a quotation within itself, in which case use double quotation marks instead of singles. The citation for the quote goes at the very end after the period in parentheses and without a period afterward, or as a footnote. The block quote should be formatted in italics, single-spaced.

> *Hermeneutics as a research methodology of sports science means that hermeneutically- oriented research is not only in the realm of scientific knowledge production as it was in the pre-scientific area, but may now be part of "empirical presumption". Hermeneutics has an important place of major scientific breadth (from medicine to philosophy) in sports science. (Haag, 1994b, S. 45)*

4.2.4 Secondary Quotations

Secondary quotations are quotes found within a primary source. They should only be used when the primary source cannot be obtained and then indicated with "cited in" within the citation. It is recommended to also include the original source in the bibliography.

Example: "In many subjects (and increasingly so), is the application of a system that makes it possible to dispense with all annotations of bibliographic data." (Eco, 1989, p. 218, cited in Chandler, 1995, p. 149).

4.2.5 Omissions and Additions

Omissions

Omissions are allowed as long as the meaning of the quotation is not changed. They are represented by dots and can be enclosed within parentheses:

[..] Or (..)	= one word
[...] Or (...)	= one sentence or a few words
[....] Or (....)	= more than one sentence

Example of original text:

"Such an ecologic perspective in macroeconomics is naturally not only connected to economic processes, but must ultimately be motivated by political processes."

Example of quotation with omission:

"Such an ecologic perspective in macroeconomics...must ultimately be motivated by political processes."

Similarly, an omission at the beginning is identified "...text" and an omission at the end of the sentence is, "text...".

Additions

Additions are supplements that are added by the author to the cited text. Citations can be supplemented if they are necessary or sensible additions.

Original text: "The comprehensive research was completed this year."

Quotation with addition: "The comprehensive research was completed this year (1987, author's note)."

4.2.6 Multiple Sources from One Author

For citing different works by a single author published in the same year (e.g. 2020), add a lower case letter after the year (e.g. 2020a) and order the works alphabetically by the title of the work.

4.2.7 Citing References from More than One Author

If more than two authors are responsible for a single source, only the first author's last name followed by "et al." is to be used; e.g. (Smith et al., 1992, p. 674).

In case of two authors, last names are separated by "&"; e.g. (Smith & Wesson, 1920, p. 77).

If citing several authors each from a different source, list each author in the order found in the bibliography including year of the publication as well as page number and separated with a semicolon.

Example: (Bortz, 1994, p. 5; Collins, 2003, p. 140; Kipling 1976, p.167, 1980, p.124; Strauss & Hague, 2010, p. 168).

4.2.8 Use of Internet Sources

Internet sources are those electronic sources, only (!) available in the world wide web. Internet sources should be used with caution because of their possible quickly changing status as well as due to a possible lack of scientific quality. The most important information to include is the URL (Uniform Resource Locator) which must be unique to the source.

Citing electronic sources within the text is be made similar to print media. In the absence of this information, try to come up with information as close to the format used for articles and books, i.e. (name, year, page number).

4.2.9 Abbreviations

All used abbreviations should be included in list of abbreviations. Domain-specific and specialist abbreviations need to be introduced in the text when used for the first time. Abbreviations used out of convenience are not allowed (e.g. "inv." for "investment" or "int'l" for "international"). When including a list of abbreviations, it shall be inserted just after the table of contents. To make it clear that the document contains an abbreviated list, a separate heading has to be added to the table of contents.

4.2.10 Tables and Figures

The purpose of tables and figures is to show findings / results (mainly data results) in a short, concise way. The origin of a table or figure must always be cited whether created by the author, copied directly or modified slightly. Each figure or table is numbered consecutively and must have a short, precise title next to the number either below or above the figure/table along with a citation of its source. All figures and tables must be listed by number in consecutive order in a separate appendix. Figures/tables generated by the research paper's author should be cited as "author's own work." When including a list of tables and/or a list of figures, it shall be inserted right after the list of abbreviations. To make it clear that the document contains a list of tables and/or figures, a separate heading has to be added to the table of contents.

4.3 Bibliography

4.3.1 General Information

In a bibliography, all references cited in the text are listed according to specific rules. Preparing a bibliography should be done with care and attention to detail.
In the bibliography, each reference cited directly or indirectly in the text (not more or less) is listed alphabetically by the author's last name with the first name shortened to its first single letter. If one source is cited multiple times in the text, it is only listed once in the bibliography.

4.3.2 Formal Layout of Bibliography

The bibliography begins on a new page as a separate section from the main content. It should be single spaced, and each second line of a source should be indented 1-1.5 cm from the edge.

The sources are included completely and have to be sorted alphabetically. Any oral source is to be listed at the end of the bibliography with the name and address of the informant. Other than this exception, no difference between different sources are made as they all belong to the same alphabetical list.

Different sources by one author are cited by the date of publication (from new to old), if more than one source per year, indicated by letters from new to old (a,b,c, etc.)

If individual and group work by one author is included in the bibliography always start with the individual work in alphabetical order first. The group work will be sorted in alphabetical order by the co-authors. Every single author has to be quoted in the bibliography. If two last names are the same, sort by first name.

Prefixes and articles are to be treated as the name (e.g. MacMillan before McCloy), whereas prepositions aren't (e.g. Persie, S. van).

Each type of source (books vs. online journal articles, ...) in a bibliography will vary in terms of content and layout required to complete the citation. The general scheme to cite in the bibliography is: Author(s), Year of Publication, Title, Publication Details.

Detailed examples for the most common sources are listed below.

- **Book with one author:** Name, Initial first name. (Year). Title. Place of Publication: Publisher.

  Lawson, J. (2008). Effective time management: How to save time and spend it wisely. London: PanBooks.

- **Book with two authors:** Name, Initial first name & Name, Inititial first name. (Year). Title. Place of Publication: Publisher.

  McCarthy, P. & Hatcher, C. (2006). *Speaking persuasively: Making the most of your presentations*. Sydney: Allen and Unwin.

- **Book with edition:** Name(s), Initial first name(s). (Year). Title. (xnd edition). Place of Publication: Publisher.

  Fisher, R., Ury, W. & Patton, B. (2007). *Getting to yes: Negotiating an agreement without giving in* (2nd edition). London: Century Business.

- **Book with Editor:** Name(s), Initial first name(s). (Editor). (Year). Title. Place of Publication: Publisher.

  Röthig, H. (Editor). (1992*). Sportwissenschaftliches Lexikon*. Schorndorf: Hofmann.

- **A chapter in a book or part of a compiled publication:** Name(s), Initial first name(s). (Year). Title of Chapter. In Initial first name(s) Name(s) (Editor). Title of Book (Pages). Place of Publication: Publisher.

  Byrne, J. (2005). Disabilities in tertiary education. In L. Rowan and J. McNamee (Editors). *Voices of a Margin* (p. 120-140). Rockhampton: CQU Press.

- **Literature without author:**

  If the author is unknown, then use the editor's name. If neither the author nor editor are known, then start with the title.

  Brofenbrenner, U. (Editor) (2005). *Making it in Marketing: communicating effectively.* Sage Publications, Thousand Oaks, California

- **Journal and newspaper articles:** Name(s), Initial first name(s). (Year). Title of Article. Name of Journal. Volume(Issue). Pages. DOI number

  Journal article

  Austin, W. (1979). Sex differences in bystander intervention in a theft. *Journal of Personality and Social Psychology.* 37(11), 2110-2120. http://doi.org/b2k6qc

  Beware of the following:

  - First name of authors are only given by the initial(s)
  - Journal title in italics
  - Between volume and issue there is no space

- Page numbers are provided without putting "p." (for page) in front

Newspaper article

Popham, B. (1987). Saving the future. *Weekend Australian Magazine*. 7-8 February. 10.

- **Part of a series:** Name(s), Initial first name(s) (Editor). (Year). Title of series. Series with No. and names. Place of publication: Publisher.

  Brehm, W., Kuhn, P., Lutter, K. & Wabel, W. (Editor). (1997). Leistung im Sport -Fitness im Leben. Beiträge zum 13. Sportwissenschaftlichen Hochschultag der Deutschen Vereinigung für Sportwissenschaft vom 22.- 24.9.1997 in Bayreuth. Band 88: Schriftenreihe der Deutschen Vereinigung für Sportwissenschaft. Hamburg: Czwalina.

- **Unpublished thesis:** Name(s), Initial first name(s). (Year). Title. Unpublished Work, Institution, Place.

  Hansen, JP. (2005). *Mechanical engineering and fuel cells.* Unpublished Ph.D thesis, University of Western Australia, Sydney.

- **Published thesis:** Name(s), Initial first name(s). (Year). Title. Unpublished Work, Institution, Place.

  Summer, B. (2007). *A survey of effective telemarketing*. Bristol UK, Canopus Publishing.

- **Source from an official institution:**

  If a source is from institutions cite the complete name, as well as if from "anonymous", cite it as anonymous..

- **Online only available journal / magazine articles:**

  This form of citation should only be used if a source is only published online. If there is also a print version available, cite according to the print versions' requirements.

  Thomas, M., Weller, V., Schulz, T. & Vörkel, C. (2001). Entwicklung einer mr-kompatiblen Schulterlagerungsschiene zur Funktionsuntersuchung der Schulter im offenen Kernspintomographen. Klinische Sportmedizin/Clinical Sports Medicine-Germany, 2(6), p. 85-93. Accessed on 23. February 2002 from http://klinische-sportmedizin.de/Auflage2001_6/ Artikel_1_Schulterschiene/Schulterschiene.pdf.

  If the source is not publicly available cite the homepage of the journal. You can also mark the end of the source with (limited access).

  Schorer, J. & Raab, M. (2001). Effekte der Teach-Back-Methode beim motorischen Lernen. Motorische Kontrolle und Lernen/Motor Control and Learning. Accessed on 22. February 2002 from http://ites.orbis-communications.de (limited access).

Internet Sources:

When citing homepages from the World-Wide-Web (www) it is necessary to give the exact dates. While quoting the date of access is necessary, it is also important to note the date of publication or last revision. The revision date is more important than the publication date. This information should be available directly on the site, in the metadata or in information about the site.

The layout for the dates is as follows: Year, Day (number). Month (written). The date of access layout: Day (number). Month (written) year.

Fromme, A., Linnenbecker, S., Thorwesten, L. & Völker, K. (1998, 19. November). Bungee-Springen aus sportmedizinischer Sicht. Accessed on 07. Februar 2002 on http://med-web.unimuenster.de/institute/spmed/ aktuell/bungee/bungee.html

4.4 Appendix

The appendix can be used for extensive materials and documents that provide further information on the topic but would interrupt the continuity if included in the text. However, the meaning of the text must not be compromised through the extension of these materials. In the main text body, there has to be a reference to the materials provided in the appendix, otherwise it is not graded.

The pages in the appendix are numbered with roman numerals and listed as such in the table of contents. Of course, there can be different headers within the Appendix – all to be referenced within the table of content.

5 Tools and Tipps for Writing an Academic Paper

There is a wide variety of useful instruments to assist academic research, writing, and project management. Below are some tools to get you started, but this is by no means an exhaustive list. Please also consult the SRH Library.

5.1 Academic Search Engines

- Google Scholar
  https://scholar.google.com/
  a replacement for Scirus, Google Scholar is a comprehensive search engine for multidisciplinary scholarly articles through Google.
- Bioline International
  http://www.bioline.org.br/
  This is a search engine with peer-reviewed academic journals on public health, food and nutritional security, food and medicine, international development issues, and biodiversity (with access to third world countries' peer-reviewed journals to boot).

5.2 Open Access and Lexica

- Directory of Open Access Journals
  https://www.doaj.org/ : DOAJ is a free academic search engine with a wide array of scientific topics (it contains over 8,000 journals on all varieties of topics). The Directory is one of the highest-valued and peer-reviewed search engines for scholars.
- EconPapers
  https://econpapers.repec.org/ EconPapers grants access to RePEc, one of the world's largest online collections of Economics working papers, journal articles, and software.

5.3 Reference Management Programms

- Citavi
  https://www.citavi.com/ A resource management program, Citavi helps organize and maintain all aspects of a major paper other than just bibliography entries.
- Endnote
  https://www.endnote.com/ A resource management program to be understood as an alternative to, Citavi for both Windows and Mac users.

- Mendeley
  https://www.mendeley.com/ A free resource to organize references and also assists with annotation/citation throughout a paper.
- BibDesk
  https://bibdesk.sourceforge.io/ A program for Mac users to organize and manage their bibliographic sources.

5.4 Other Useful Sources

- Phrasebank
  https://www.phrasebank.manchester.ac.uk/ A phraseological resource for academic writers organised according to the main sections of a research paper or dissertation.
- Mind42
  https://mind42.com/ A free mind mapping tool online, Mind42 is an excellent way to map out brainstorming ideas and save them electronically, it does mandate creating an account to save/share work.

5.5 Final Tipps

Consider the following practical tips when writing an academic paper:

- Work in a systematic manner. The reader should be able to easily understand the course of investigation.
- Work from the generic to the specific → deductive approach.
- Link each section to the ones before.
- Own interpretations of findings are essential, but the logic of argumentation has to be clear. Every step of argumentation needs to be presented clearly and transparently. Supporting arguments from other sources should be used but must be marked and clearly separated from the author's own opinion.
- The more high(est) quality references, the higher the quality of the document.
- Clean and well-structured content is effective when conveyed in the highest linguistic sense (see Chapter 3).
- Keeping in line with the formalities of academic work (see Chapter 4) is a basic requirement for any scientific research paper.

6 Evaluation Criteria for Academic Research

This chapter outlines the most common and important evaluation criteria for scientific documentation, like (Bachelor and Master) theses, seminar papers, and internship reports. It lists major differences and requirements for a variety of needs. The purpose of this chapter, however, is not to provide an all inclusive list for any sort of scientific text but to provide an overall understanding of some of the most important evaluation criteria within the scientific community. Hence, it remains the responsibility of the author of any scientific text to assure all criteria are met.

6.1 Guidance on the qualitative differences between Bachelor and Master theses

- For Bachelor theses, it is much more acceptable to summarize (especially so in chapter 2 of a thesis); whereas on Master level paraphrasing und synthesizing becomes much more important.
- For Bachelor theses, it is fine to simply state rather briefly the way data has been collected and analyzed in chapter 3; whereas on Master level chapter 3 is an elaborate account of pros and cons of the chosen methodological design as well as tools to collect and analyze data.
- For Master theses, it is important to provide a change of perspectives on key topics and themes in order to demonstrate critical reflection and critical thinking.
- For Master theses, it is more important to use theories and models to guide data collection and data analysis.
- For Master theses, it is much more important to stick to high rated academic/scientific sources of references, mainly derived from A-rated journal articles; whereas on Bachelor level more C-rated sources as well as books are acceptable.

6.2 Assessment Criteria for Content and Structure

- An abstract is required only for theses (not term papers or internship reports). It is a short version of the academic paper and includes a brief summary of the research question / objective, scope, methodology used, main findings and interpretation.
- A coherent outline showing the ability of the author to structure and analyze a topic. The outline should provide an overview regarding the logical structure of the document and informs about the correct weighting of the individual parts of a work.

- Stringent description of a practical phenomenon / problem indicating the relevance and importance of solving the problem. Thereof derived research question (indication of several, often up to three related and reliable central questions). The research question should be explicitly formulated as such.
- Stringent description of the course of investigation and methodology approach for solving the research question. If an empirical approach is used to answer the central research question, then first the theoretical framework should be explained before diving into the methodological aspects of data analysis.
- A thorough review of the existing literature within the field of study (for seminar papers and internship reports this is not required but can positively enhance the work). Discussing recent research findings and emphasizing research gaps is to prove and distinguish that the research question presented is justified from a theoretical perspective.
- Development of a theoretical framework on which a rigorous, systematic, and (ideally) fully integrated answer to the central research question can be based. The framework itself can be based on several methodologies:
    - one or more existing theories
    - causal logic considerations
    - or a combination of existing theories with causal theories

  In addition to rigorous and systematic aspects, its quality is also measured by the sufficiency of literature referenced and the quality of the sources (e.g. A-rated journals).
- Depth of analysis / discussion of main findings. The central research question is answered based on
    - existing results and their interpretation in terms of the central research question
    - empirical data collection, analysis and interpretation
    - or a combination of the aforementioned aspects
- The integration of an empirical study is not required to ensure a good grade. More importantly, its function is relative to the central question. The empirical approach must be derived from the theoretical framework in place, especially with quantitative and/or qualitative nature of the data being essential to the central question. When evaluating the data, usually a purely descriptive level of consideration should be abandoned for suitable methods of inferential statistics. Internship reports and term papers are more likely to do without primary data collection.

- In theses and term papers, the contributions to fill out (or develop) the theoretical framework by the author must be clearly identified. For seminar papers, depending on the topic, a purely descriptive presentation and discussion of existing knowledge can be sufficient.
- Final summary of the findings and answering the central research question in form of precise key messages. In this context, it is also necessary to point out the limitations of the findings, i.e. to show a critical reflection of one's work. Furthermore, the outlook of future research to be expected in the field must be given to complete the conclusion.
- Usage of high-class literature, mainly A-rated journal articles throughout all chapter, with a slight exception granted for chapter 1, where references might include a higher portion of lower-standard literature.

6.3 Language Evaluation Criteria

The basis for the assessment of language is defined by this guideline's formulation rules as stated in Chapter 3.

6.4 Formal Evaluation Criteria

The formal criteria used to evaluate a research paper are to comply with the requirements for:

- text design and layout
- citations and references (which includes to provide page numbers)
- appendices and any supplemental information as defined by this set of guidelines.

The dedicated compliance with the formal requirements as stated herein is a prerequisite for good and excellent marks of papers. Submitted work that does not meet the formal requirements may not receive a grade above a satisfactory level.

6.5 Scope of Academic Papers

In principal, the grade of assessment is not based on scope and breadth of the work, but instead rests particularly on the structure, content, language level, and form within. However, certain benchmarks are useful to measure what a "reasonable" amount of work implies. Hence, the following guidelines are provided (subject to your supervisor´s discretion):

- Bachelor Thesis: 50-60 pages of content
- Master Thesis: 60-80 pages of content
- Internship Reports: >15 pages of content (mandatory)

In this respect, "content" is referred to as anything from the start of chapter 1 until the end of chapter 5. Hence, it does not include any lists (of abbreviations / diagrams / etc.) or appendices nor references provided.

6.6 Distinctions of Evaluation Criteria

In addition to the already-mentioned differences among theses, internship reports, and seminar papers there are distinctions in their evaluations:

- At the Master-level, evaluation criteria are different from the Bachelor requirements concerning the intellectual depth in answering the central research question. The substance of content must be significantly more pronounced at the Master level. Quality requirements for structural coherence and language level can be at lower level for Bachelor work. The formal requirements, however, remain identical. Especially Chapter 2 and 3 of the typical 5-Chapter-Thesis-Structure shall clearly show the elevated level of scientific writing of a Master. All chapters shall include more than just a summary but use paraphrasing and critical thinking (e.g. inclusion of different perspectives). Finally, the references provided for a Master thesis have to be of higher quality (e.g. more A-rated journals).
- Internship reports are theses with reduced scope. They require the same quality criteria as theses.
- Seminar papers fall under the same quality standards as theses and internship reports. The rules for structure and language level as well as the formal requirements are identical.

7 FINAL NOTE

As already mentioned in the Preface, any scientific document can to be understood as a business card: It is a chance to shine, a first but yet long-lasting impression which follows certain standards. The less experienced one is, the more important it becomes to be aware and adhere to any such standards. This document is meant to assist, provide important guidelines. Nevertheless, it is the responsibility of each author not to blindly adhere to the guidelines presented herein, but to (pro-) actively assure standards of the scientific community are met above and beyond this guideline.

Just like a business card, a scientific document is something presented to others and even when it one day in the future might represent nothing but the past, it might still reflect the present for somebody else. So make sure to be proud of it above and beyond mere requirements laid open herein.

In case of any doubts or questions, feel free to get in touch with Prof. Dr. Carsten Held.